

Multiple Choice Answers:

Year	Question	Answer
2014	4	D
	18	A
	23	A
2013	8	D
2012	2	C
	4	B
	6	C
	22	C
	23	D
2011	4	C
	5	B
2010	9	B
	10	A

2014 WACE EXAM:

Question 28

(6 marks)

For the species listed in the table below, draw Lewis structure (electron dot) diagrams.

All valence shell electron pairs should be represented either as : or as —

(for example, water $\text{H}:\ddot{\text{O}}:\text{H}$ or $\text{H}-\ddot{\text{O}}-\text{H}$ or $\text{H}-\overline{\text{O}}-\text{H}$)

Description	Marks
$[\text{Ca}]^{2+}$ $[\text{:}\ddot{\text{O}}\text{:}]^{2-}$ - Ca shown with no valence electrons - O shown with 8 valence electrons - Ions in brackets with correct charges	1 1 1
$\begin{array}{c} \text{:}\ddot{\text{O}}\text{—}\ddot{\text{Cl}}\text{:} \\ \\ \text{:}\ddot{\text{Cl}}\text{:} \end{array}$ 1 bonding pair or covalent bond shown between O and each Cl atom 2 non-bonding pairs of electrons shown on the oxygen 3 non-bonding pairs of electrons shown on both Cl atoms or $\text{:}\ddot{\text{Cl}}\text{—}\ddot{\text{Cl}}\text{—}\ddot{\text{O}}\text{:}$ 2 bonding pairs or covalent bonds one each between middle Cl and the other two atoms 2 non-bonding pairs of electrons on the middle Cl atom shown on the oxygen 3 non-bonding pairs of electrons shown on outer Cl atom and O atom	1 1 1 or 1 1 1
Total	6

Question 30
(8 marks)

- (a) Classify the following pure substances into their bonding class by placing each symbol or formula into the appropriate box in the table below. (6 marks)

Fe Si NH_4Cl HCl H_2O CaCl_2

Metallic	Ionic	Covalent network	Covalent molecular
Fe	NH_4Cl CaCl_2	Si	HCl H_2O

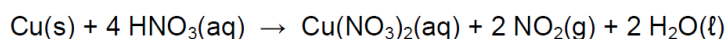
Description	Marks
One mark for each correct answer No marks allocated to a formula used more than once.	1–6
Total	6

- (b) What is it about the bonding of solid metals that explains why they are good conductors of electricity? (2 marks)

Description	Marks
• Their valence electron are delocalised	1
• Valence electrons are free to move through the solid lattice of cations, carrying their negative charge	1
Total	2

Question 31
(12 marks)

- (c) When concentrated nitric acid is added to copper metal, a brown gas, NO_2 , is produced. The equation is shown below.



0.444 g of Cu reacts with excess 6.00 mol L^{-1} nitric acid.

- (i) Calculate the number of moles of Cu consumed. (2 marks)

Description	Marks
$n(\text{Cu}) = m/M = 0.444/63.55$	1
$= 0.0069867 = 6.99 \times 10^{-3} \text{ mol}$	1
Total	2
NB: No penalty if units missing or more than 3 significant figures are given.	

- (ii) Calculate the number of moles of NO_2 produced. (2 marks)

Description	Marks
$n(\text{NO}_2) = 2 \times n(\text{Cu}) = 2 \times 0.0069867$ or $2 \times 6.99 \times 10^{-3}$ or 2 x value given in (c)(i)	1
$= 0.013973 = 1.40 \times 10^{-2} \text{ mol}$	1
Total	2
NB: No penalty if units missing or more than 3 significant figures are given	

2013 WACE EXAM:**Question 28****(4 marks)**

For the species listed in the table below, draw electron dot diagrams.

All valence shell electron pairs should be represented either as : or as —

(for example, water $\text{H}:\ddot{\text{O}}:\text{H}$ or $\text{H}-\ddot{\text{O}}-\text{H}$ or $\text{H}-\overline{\text{O}}-\text{H}$)

Species	Electron dot diagram
Na_2S	$2 [\text{Na}]^+ \quad [:\ddot{\text{S}}:]^{2-}$
SO_2	$:\ddot{\text{O}}-\ddot{\text{S}}=\ddot{\text{O}}:$

Description	Marks
Max 2 marks each	1–4
If non-bonding electrons are not shown in one or both answers, award a maximum of 3 out of 4 marks	
Resonance structures for SO_2 acceptable	
Total	4

Question 31**(10 marks)**

- (a) List **two** features of heterogeneous mixtures that make them different from homogeneous mixtures. (2 marks)

Description	Marks
Non uniform composition throughout the mixture	1
Non uniform properties throughout the mixture	1
Total	2

- (b) Complete the table with one example of each category to illustrate the difference between elements, compounds and mixtures. (4 marks)

Element	Compound	Heterogeneous mixture	Solution

Description	Marks
One mark per category if every example per category is correct	1–4
Note: Air (a sample of air) is a solution; the air/the atmosphere is heterogeneous Salt water is a solution; the ocean is heterogeneous	
Total	4

- (c) In terms of electron transfer, explain how the atoms of magnesium and bromine can react together to form the ionic salt, magnesium bromide. (4 marks)

Description	Marks
• Electrons are transferred from the magnesium atoms to the bromine atoms (as evident by a reaction)	1
• (In this process,) magnesium atoms lose electrons to become positively charged while	1
• bromine atoms gain electrons to become negatively charged ions	1
• (Because of the mutual) attraction of the positive and negative ions, the magnesium and bromine ions surround each other to form the (crystalline) lattice of the (ionic) salt	1
Total	4

Question 32
(18 marks)

The physical properties of substances are due to their bonding and structure.

- (a) Complete the following table, outlining the differences in the physical properties of calcium fluoride and fluorine. (6 marks)

	calcium fluoride	fluorine
Type of bonding	ionic	covalent
Electrical conductivity as a solid (high, low, nil)	nil	nil
Electrical conductivity as an aqueous solution (high, low, nil)	high	Nil or low

Description	Marks
1 mark per correct answer	1–6
Total	6

- (b) (i) Does calcium fluoride or fluorine have the higher melting point? Circle your answer (1 mark)

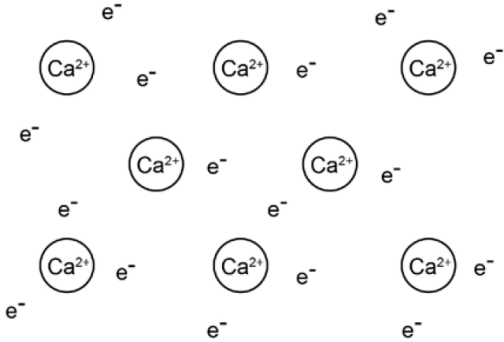
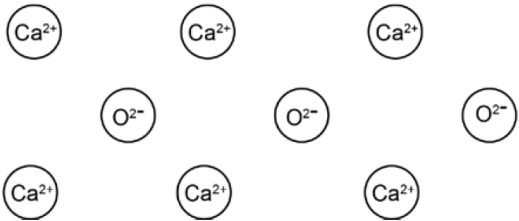
calcium fluoride

fluorine

- (ii) By describing the bonding present in each substance, explain the difference in melting points. (5 marks)

	Description	Marks
(i)	calcium fluoride	1
(ii)	• calcium fluoride is held together in an ionic lattice by strong ionic bonding (each ion surrounded by and bonded strongly to several ions of opposite charge)	1
	• requiring large amounts of energy to overcome these attractive forces and so melt	1
	• fluorine molecules are only loosely held together by relatively weak dispersion forces that	1
	• require relatively low amounts of energy to overcome and so melt	1
	If not otherwise mentioned: • The stronger the forces of attraction holding the particles of the substance together, • the more energy is required to overcome these forces of attraction resulting in • the higher the temperature at which it melts	1
Total		6

- (c) (i) Use labelled diagrams to show the bonding structure of solid calcium and solid calcium oxide. (4 marks)

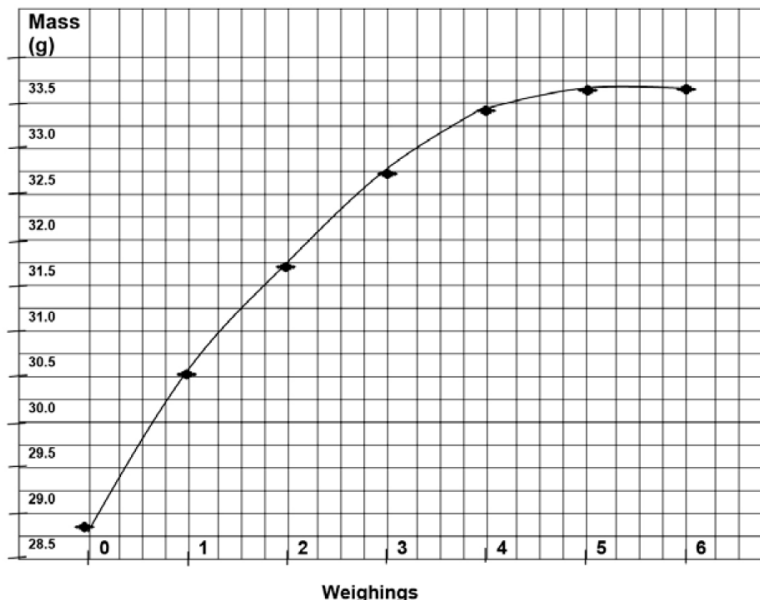
Description	Marks
Calcium 	1–2
Calcium oxide 	1–2
Total	4

- (ii) Describe how this bonding structure affects each substance's electrical conductivity. (2 marks)

Description	Marks
Calcium: Delocalised electrons able to transfer charge by moving around the metallic lattice.	1
Calcium oxide: No mobile charge carriers – valence electrons are localised on the ions.	1
Total	2

Question 38

(29 marks)

	Description	Marks																
(a)	One mark each for any two different reasonable hazards: eg • Burns from Bunsen burner • Burns from hot tripod, gauze mat, crucible or lid • Inhalation of white unknown fumes • Ignition of magnesium ribbon and subsequent eye damage	1–2																
(b)	One mark each for any two different precautions: eg • Use of eye protection • Use of metal tongs to pick up crucible • Use of fireproof mat • Conduct inside a fume cupboard	1–2																
(c)	To ensure that the magnesium strip was clean and pure, free from any corrosion (or magnesium oxide)	1																
(d)	To ensure than any reaction that occurred had gone to completion.	1																
(e)	To prevent any contamination into the reaction or to avoid any product escaping as white smoke Magnesium oxide in powder form easily lost	1																
(f)	Any 2 of • Supports the crucible above the Bunsen burner flame • allows direct heating • stops it toppling over	1–2																
(g)	One mark each for: • Labelled axes • Appropriate scale • Accuracy of points • Smooth curve (line) of best fit  <table><caption>Data points from the graph</caption><thead><tr><th>Weighings</th><th>Mass (g)</th></tr></thead><tbody><tr><td>0.0</td><td>28.8</td></tr><tr><td>0.8</td><td>30.4</td></tr><tr><td>1.8</td><td>31.6</td></tr><tr><td>2.8</td><td>32.6</td></tr><tr><td>3.8</td><td>33.2</td></tr><tr><td>4.8</td><td>33.4</td></tr><tr><td>5.8</td><td>33.4</td></tr></tbody></table>	Weighings	Mass (g)	0.0	28.8	0.8	30.4	1.8	31.6	2.8	32.6	3.8	33.2	4.8	33.4	5.8	33.4	1–4
Weighings	Mass (g)																	
0.0	28.8																	
0.8	30.4																	
1.8	31.6																	
2.8	32.6																	
3.8	33.2																	
4.8	33.4																	
5.8	33.4																	

(h)	$m(\text{Mg}) = 28.75 - 21.25 = 7.50 \text{ g}$	1
(i)	$n(\text{Mg})_{\text{initial}} = 7.50 / 24.31$ $= 0.308515$ $= 0.309 \text{ mol}$	1 1
(j)	$n(\text{Mg})_{\text{final}} = 0.309 \text{ mol}$ (no atoms lost in the reaction)	1
(k)	$\Delta m = 33.65 - 28.75 = 4.90 \text{ g}$	1
(l)	Oxygen had been added to the magnesium to form magnesium oxide	1
(m)	$n(\text{O}) = 4.90 / 16.00$	1
	$= 0.30625$ $= 0.306 \text{ mol}$	1
(n)	$n(\text{Mg}) : n(\text{O})$ $0.309 : 0.306$ $1 : 1$	1 1
(o)	MgO	1–2
(p)	1 mark each for any two reasonable sources given: eg • Loss of MgO when lifting crucible lid • Reaction not yet gone to completion • Not all corrosion / dirt removed from magnesium ribbon • Inaccuracy of measuring mass • Tolerance of mass balance • Rounding error	1–2
(q)	$2 \text{ Mg(s)} + \text{O}_2(\text{g}) \rightarrow 2 \text{ MgO(s)}$ 1 mark for correct formulas and balanced 1 mark for correct state symbols (Allow follow through marks from formula used in (o))	1–2
	Total	29

2012 WACE EXAM:

Question 28

(4 marks)

For the species listed in the table below, draw electron dot diagrams.

Species	Electron Dot Diagram
CaCl_2	$\left[\text{Ca} \right]^{2+} \quad 2 \left[\begin{array}{c} \cdot\cdot \\ :\text{Cl}: \\ \cdot\cdot \end{array} \right]^{-}$
PCl_3	$ \begin{array}{c} :\ddot{\text{Cl}} - \ddot{\text{P}} - \ddot{\text{Cl}}: \\ \\ :\ddot{\text{Cl}}: \end{array} $

Description	Marks
Max. 2 marks each	1–4
If non-bonding electrons are not shown in one or both answers, award maximum 3 out of 4.	
Total	4

Question 32**(9 marks)**

- (a) For each of these two substances, state the type of bonding, a use and the property of each substance that allows it to be used as suggested. (6 marks)

Substance	Type of bonding	Use	Property
graphite	Covalent network	• 'Lead' pencils • Dry lubricant • High temperature moulds and crucibles • Electrodes • Fire retardant	• Soft • Weakly held layers that slide over each other • Chemically resistant • High melting point • Good conductor of heat • Good conductor of electricity
copper	Metallic	• Gas pipes • Water pipes • Electrical wires • Electronic circuit boards • Making alloys (brass & bronze) • Coins • Art • Heat sinks • Water drains • Cookware	• Malleable • Ductile • High lustre when polished • Very good conductor of heat • Very good conductor of electricity • Resistant to corrosion

Description		Marks
Graphite	Covalent network bonding	1
	A use (see above for examples)	1
	A property that is clearly linked to the use stated (see above)	1
Copper	Metallic bonding	1
	A use (see above for examples)	1
	A property that is clearly linked to the use stated (see above)	1
Total		6

- (b) Describe how the theory of bonding in copper can explain the property stated in Part (a). (3 marks)

Description	Marks
One property previously mentioned	1–3
Clear relevant explanation of copper's bonding that gives rise to the property stated, and explain the link	
Total	3

2011 WACE EXAM:
Question 28
(8 marks)

- (a) For the species listed in the table below, draw electron dot diagrams. (4 marks)

All valence shell electron pairs should be represented either as : or as —

(for example, water $\text{H}:\ddot{\text{O}}:\text{H}$ or $\text{H}-\ddot{\text{O}}-\text{H}$ or $\text{H}-\overline{\text{O}}-\text{H}$)

Species	Electron Dot Diagram
K_2O	$2 \left[\text{K} \right]^+ \quad \left[\ddot{\text{O}} \right]^{2-}$
NCl_3	$ \begin{array}{c} :\ddot{\text{Cl}}: \quad \text{---} \quad \ddot{\text{N}}: \quad \text{---} \quad :\ddot{\text{Cl}}: \\ \\ :\ddot{\text{Cl}}: \end{array} $

Description	Marks
2 marks each	1–4
Question incorrectly answered	0
Question not attempted	–
Cumulative Total	4

Note: Deduct 1 mark overall if non-bonding electrons not shown in one **or** both answers

- (b) Diamond and methane contain covalently bonded carbon atoms, yet their melting points are very different (3547°C and -182°C respectively). State the reasons for the difference in melting points in terms of chemical bonding. (4 marks)

Description	Marks
Diamond: Covalent network bonding.	1
Methane: Intermolecular forces between molecules.	1
Covalent network bonding in diamond is much stronger than weak intermolecular forces in methane.	1
More energy (heat) required to disrupt covalent network bonding diamond than that required to disrupt weak intermolecular forces in methane. Therefore, melting point of diamond is significantly higher than that of methane.	1
Question incorrectly answered	0
Question not attempted	–
Cumulative Total	4

2010 WACE EXAM:**Question 28****(4 marks)**

For each species listed in the table below, draw the structural formula, representing all valence shell electron pairs either as : or as —.

(for example, water $\text{H}:\ddot{\text{O}}:\text{H}$ or $\text{H}-\ddot{\text{O}}-\text{H}$ or $\text{H}-\overline{\text{O}}-\text{H}$)

Description		Marks
Carbon dioxide (CO ₂)	$\ddot{\text{O}}=\text{C}=\ddot{\text{O}}$	1–2
Magnesium fluoride (MgF ₂)	$[\text{Mg}]^{2+} \quad 2 \quad [:\ddot{\text{F}}:]^{-}$	1–2
-1 for non-bonding electrons missing -1 for brackets/charge missing -1 for each other error		
		Total 4

Question 42**(9 marks)**

Sodium carbonate, carbon dioxide and silicon dioxide (sand) are compounds that contain oxygen. However, their physical properties are quite different. A selection of these properties is given on the table below.

Compound	Melting point (°C)	Electrical conductivity as a solid	Electrical conductivity as a liquid
Na ₂ CO ₃	851	Very poor	Good
SiO ₂	1713	Very poor	Very poor
CO ₂	-57	Very poor	Very poor

Explain these properties of each substance in terms of their bonding.

Description	Marks
3 marks per compound – 1 mark per property. The following points should be made: Sodium carbonate - High melting point explained by strong force of attraction between positive and negative ions. Large amount of energy required to break bonds. - Non-conductive as a solid explained by ions in fixed positions in a lattice. No mobile charge, no conduction/ - Conducts as a liquid explained by melting allowing the ions to move freely and so conduct electricity. (Must be specific about the presence of ions for full marks)	1–3
Silicon dioxide - High melting point explained by presence of strong covalent bonds between all atoms (shared electrons). Large amount of energy required to break bonds. - Non-conduction as a solid as the particles in the lattice are neutral atoms – no mobile charge. - Non-conduction as a liquid – atoms are able to move but are neutral – no mobile charge. (Must be specific about the presence of atoms)	1–3
Carbon dioxide - Low melting point explained by presence of weak intermolecular forces between molecules (strong covalent bonds between atoms). Small amount of energy required to break bonds. - Non-conduction as a solid explained by molecules being neutral. No mobile charge carriers → no conduction of electricity. - Non-conduction as a liquid explained by molecules being neutral. No mobile charge → no conduction of electricity. (Must be specific about the presence of molecules)	1–3
	Total 9